Virtual brush

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- 1. Virtual brush
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MediaPipe is a cross-platform machine learning framework that can process key point detection such as hands, faces, and postures in videos and images in real time.

1. Virtual brush

MediaPipe Virtual brush can track finger status through hand detection and switch different modes according to finger status:

When the index finger and middle finger of the right hand are combined, it is the selection state, and the color selection box pops up at the same time. When the two fingertips move to the corresponding color position, the color is selected (black is the eraser);

When the index finger and middle finger are separated, it is the drawing state, and you can draw anything on the drawing board.

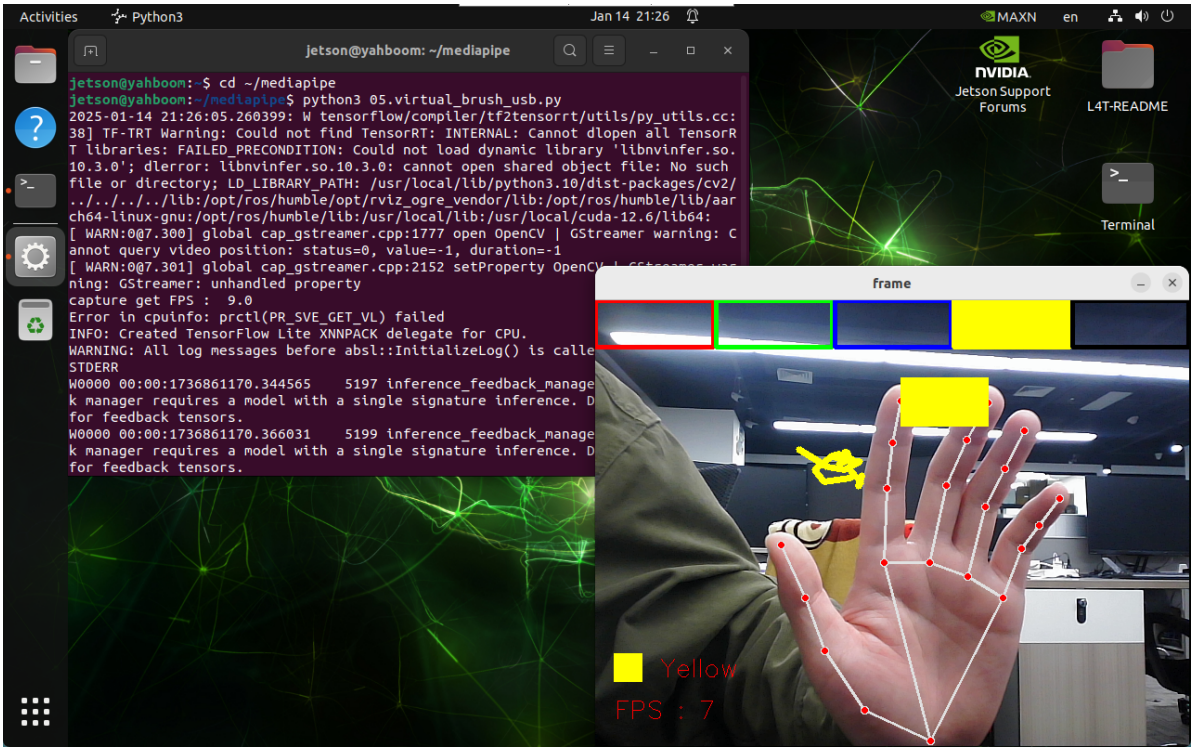
2. Effect preview

2.1. USB camera

```
cd ~/mediapipe
```

```
python3 05.virtual_brush_usb.py
```

Note: Select the camera screen and press `q` to exit the program!


Activities
Python3
Jan 14 21:26
MAXN
en
jetson@yahboom: ~/mediapipe
jetson@yahboom:~$ cd ~/mediapipe
jetson@yahboom:~/mediapipe$ python3 05.virtual_brush_usb.py
2025-01-14 21:26:05.260399: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Could not find TensorRT: INTERNAL: Cannot dlopen all TensorRT libraries: FAILED_PRECONDITION: Could not load dynamic library 'libnvinfer.so.10.3.0'; dlerror: libnvinfer.so.10.3.0: cannot open shared object file: No such file or directory; LD_LIBRARY_PATH: /usr/local/lib/python3.10/dist-packages/cv2/../../../../lib:/opt/ros/humble/opt/rviz_ogre_vendor/lib:/opt/ros/humble/lib/aarch64-linux-gnu:/opt/ros/humble/lib:/usr/local/lib:/usr/local/cuda-12.6/lib64:
[WARN:0@7.300] global cap_gstreamer.cpp:1777 open OpenCV | GStreamer warning: Cannot query video position: status=0, value=-1, duration=-1
[WARN:0@7.301] global cap_gstreamer.cpp:2152 setProperty OpenCV | GStreamer warning: GStreamer: unhandled property
capture get FPS : 9.0
Error in cpuinfo: prctl(PR_SVE_GET_VL) failed
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
WARNING: All log messages before absl::InitializeLog() is calle
STDERR
W0000 00:00:1736861170.344565 5197 inference_feedback_manage
k manager requires a model with a single signature inference. D
for feedback tensors.
W0000 00:00:1736861170.366031 5199 inference_feedback_manage
k manager requires a model with a single signature inference. D
for feedback tensors.
NVIDIA
Jetson Support Forums
L4T-README
Terminal
frame
Yellow
FPS : 7